AADITA NAG

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EDUCATION

B.E. CSE, SVIT Bengaluru (CGPA: 9.4/10, '27) | **B.S. Data Science, IIT Madras** (CGPA: 8.0/10, '27) | **XII CBSE, Don Bosco** (92%, '21) **| X CBSE, St. Mary's** (96%, '19)

TECHNICAL SKILLS

Languages: C, C++, Java, Python, JavaScript (ES6+), Kotlin
Web: HTML5, CSS3, React.js, Vue.js, Tailwind CSS
Backend & Databases: Node.js, Express.js, Flask, REST APIs, MongoDB, PostgreSQL, SQLite, SQL, Mongoose
AI/ML & Computer Vision: YOLOv8, OpenCV, EasyOCR, PyTorch, TensorFlow
Tools & Platforms: Git, GitHub, Postman, VS Code, Vercel, Render, n8n

EXPERIENCE

Software Engineer Intern - Sthavishtah (VTU-registered startup) *Jan '26 - Aug '26*

- Contributed to the end-to-end development and maintenance of the company website, collaborating with a 5-member team across 2 projects: an Intelligent Health Record & Cybersecurity Platform and a Raspberry Pi-based hardware system.
- Developed an AI-powered Yoga Pose Analyser to analyse yoga postures, achieving 96% accuracy.
- Supported programs and events that reached approximately 200 participants per event and an online program with 50 active participants.

LEADERSHIP & POSITIONS OF RESPONSIBILITY

Super-Coordinator, Central Events (Paradox) - IIT Madras *Jan '26 - Jul '26*

- Lead campus-wide ambience planning for Paradox Fest 2026, directing a 25-person team along with outreach.

Web Admin, Pichavaram House - IIT Madras *Aug '25 - Aug '26*

- Head the WebOps Group; oversee 5 subdivisions and all live websites and virtual meeting infrastructure for the house.

Treasurer, Coding Club - SVIT, Bangalore *Feb '26 - Present*

- Designed promotional materials in Canva and hosted coding contests on HackerRank with 200+ participants

Coordinator, Central Events (Paradox) - IIT Madras *Sep '24 - Jun '25*

- Managed flagship fest events for Paradox 2025 - DJ Night, Comedy Night, and opening/closing ceremonies.

ACADEMIC PROJECTS

AI Traffic Violation Detection with e-Challan System *GitHub Link*

- Engineered a computer vision pipeline using **YOLOv8** and **OpenCV** to detect 3 traffic violations: helmet violations, triple riding, and red-light violations.
- Integrated **EasyOCR-based** license plate recognition with **Flask** and **MongoDB** to link detected violations to vehicle records and automate e-challan generation and penalty calculation.
- Implemented **Twilio-based** SMS notifications and a **React** dashboard for violation history, challan management, and reporting; configured helmet-model training for 50 epochs with 640 × 640 images.

ML-Driven Smart Helmet with Predictive Impact Airbag System *GitHub Link*

- Worked on a real-time crash-detection pipeline integrating **IMU telemetry**, **Raspberry Pi** edge inference, and **GPIO-controlled** safety actuators for rapid crash response.
- Built a **Three.js 3D** cockpit simulation with **WebSocket-based** sensor streaming and crash scenarios to evaluate model detection latency independently of manual triggers.
- Validated event routing, duplicate-event handling, and actuator safety interlocks using automated Pytest suites, with hardware deployment safeguards to prevent unintended airbag activation.

ACHIEVEMENTS

- **GO-BRICS** International Hackathon Semi-finalist & **Social Media Manager/Content Lead** - Top 69 of 12,000+ teams
- **3× Hackathon Winner**; 1st Place, **Mini Project Exhibition**; 2nd Place, **Web Craft Design**.
- IIT Madras: 2nd Place, Settlement Design Challenge 3.0; 3rd Place, 2.0; 2nd Place, Twist and Turn.